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README: Computational Verification Code for
"On the Maximum Algebraic Degree of APN Functions
via Partial APN Functions and Shifted Derivative Images"
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File: apn_computations.py

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OVERVIEW

This script verifies all computational claims in the paper using only the Python 3 standard library. No numpy, sage, or third-party packages are required.

All section headings in the code and in this README use descriptive mathematical names rather than theorem/proposition numbers. If the paper is renumbered, the code and README remain correct without any changes.

The paper studies the conjecture (Budaghyan–Carlet–Helleseth–Li–Sun, 2018) that no APN function over F_{2^n} has algebraic degree n for $n \geq 3$, and reformulates it as: the projective line-labeling

$$\lambda_F(\{a, b, a+b\}) = F(a) + F(b) + F(a+b)$$

must be surjective onto $F_{2^n}^*$ for every APN F with $F(0)=0$.

The script verifies the structural and computational results supporting this reformulation.

USAGE

Fast verifications (a few seconds):

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python3 apn_computations.py
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Slow verifications (includes exhaustive GF(8) check ~30-60 seconds and monomial surjectivity checks for $n=4,5,6,7$):

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python3 apn_computations.py --slow
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Python version: 3.7 or later. No external packages needed.

WHAT THE SCRIPT VERIFIES

SECTION 1: 0-APN COUNTEREXAMPLE

(properness alone does not force surjectivity)

Mathematical claim:

There exists $F: \text{GF}(16) \rightarrow \text{GF}(16)$ with $F(0)=0$ that is 0-APN

(no projective line receives label 0) but not APN (some derivative

$D_a F$ is not 2-to-1), and whose line-labeling λ_F is not surjective.

Why this matters:

A proper nonzero line-labeling (= 0-APN-ness) is not sufficient for

surjectivity. The full APN condition -- every $D_a F$ is 2-to-1, i.e.,

every two intersecting lines receive distinct labels -- is genuinely

needed. This shows the surjectivity conjecture captures exactly the

right strength of the APN hypothesis.

Explicit counterexample verified:

$F = [0, 5, 7, 5, 14, 12, 8, 7, 10, 1, 7, 1, 10, 12, 8, 2]$

over $GF(16) = GF(2)[\alpha]/(\alpha^4 + \alpha + 1)$,
elements encoded as integers 0..15 via binary
representation.

Output:

F is 0-APN: True
F is APN: False
Missing labels: [2, 4, 6, 8]

SECTION 2: EXHAUSTIVE SURJECTIVITY CHECK OVER $GF(8)$ [--
slow]

Mathematical claim:

Every APN function $F: GF(8) \rightarrow GF(8)$ with $F(0)=0$ gives a
surjective
line-labeling λ_F .

Background:

For $n=3$, $PG(2,2)$ is a single Fano plane with 7 lines.

APN properness

forces all 7 labels to be pairwise distinct and nonzero
(intersecting

lines have different labels, no line has label 0), so all
7 nonzero

field elements appear automatically. The exhaustive
check confirms

this for every APN function.

Method:

All $8^7 = 2,097,152$ functions with $F(0)=0$ are enumerated.

Each is tested for the APN property, then for
surjectivity.

Expected output:

Total functions with $F(0)=0$:	2,097,152	(= 8^7)
APN functions:	86,016	
Surjective line-labeling:	86,016	
Non-surjective:	0	

Timing: approximately 30-60 seconds.

SECTION 3: APN MONOMIAL SURJECTIVITY CHECK [--slow] (n = 4, 5, 6, 7)

Mathematical claim:

For n = 4, 5, 6, 7, every APN monomial $F(x) = x^d$ over $GF(2^n)$
gives a surjective line-labeling $\lambda_{F, \cdot}$.

Method:

All exponents d in $[2, 2^n-2]$ are tested for the APN property.

Canonical representatives are reported (d and 2^n-1-d give the

same function up to the substitution $x \rightarrow x^{-1}$, which is

CCZ-equivalent and preserves APN-ness).

For each APN monomial, the line labels are computed and compared
to $F_{2^n}^*$.

The $GF(2^n)$ irreducible polynomials used are:

n=4:	$x^4 + x + 1$	(integer form: 19)
n=5:	$x^5 + x^2 + 1$	(integer form: 37)
n=6:	$x^6 + x + 1$	(integer form: 67)
n=7:	$x^7 + x + 1$	(integer form: 131)

Expected output (APN monomial counts):

n=4: 4 monomials (canonical exponents [3, 6])

n=5: 25 monomials

n=6: 6 monomials (canonical [3, 6, 12, 15, 24, 30])

n=7: 77 monomials

All surjective.

SECTION 4: S₃-ORBIT DIVISIBILITY $6 \mid N_b(F)$ for every $b \neq 0$

Mathematical claim:

For every APN function F with $F(0)=0$ and every nonzero b,
the count $N_b(F) = |\{(x,y) : F(x)+F(y)+F(x+y) = b\}|$ is

divisible by 6.

Why:

$N_b(F) = 6 * L_b(F)$ where $L_b(F)$ is the number of projective lines with label b . Each unordered line contributes exactly 6 ordered pairs (x,y) : the group $S_3 = GL(2,2)$ of order 6 acts freely on nonzero second-order fibers, permuting $\{x, y, x+y\}$.

This corrects the tempting but false mod-4 argument. The symmetry group has order 6, not 4. The total nonzero mass

$(2^n-1)(2^{n-2})$ is also divisible by 6, so no congruence contradiction arises from $N_1(F) = 0$.

Dimensions tested: $n = 3, 4, 5, 6, 7$ (using Gold $F(x) = x^3$).

SECTION 5: OVERSIZED-FIBER OBSTRUCTION AND THRESHOLD U_n

Mathematical claim (the obstruction):

If $N_1(F) = 0$, then some nonzero $b \neq 1$ satisfies $N_b(F) \geq U_n$,

where $U_n = 6 * \text{ceil}(2^n / 6)$.

$U_n = 2^n + 2$ for even n (since $2^n \equiv 4 \pmod{6}$)

$U_n = 2^n + 4$ for odd n (since $2^n \equiv 2 \pmod{6}$)

Contrapositive (the usable direction):

If all $N_b(F) < U_n$, then $N_1(F) > 0$, so $F+x^{2^n-1}$ is not APN.

Proof sketch:

If $N_1(F)=0$ then $\sum_{b \neq 0,1} N_b(F) = (2^n-1)(2^{n-2})$, distributed over 2^{n-2} terms with average 2^{n-1} . Since each

term is a positive multiple of 6, none can equal the odd value

2^{n-1} , so some term exceeds 2^{n-1} . The next multiple of 6

above 2^{n-1} is U_n .

The script also prints the threshold table for $n = 3, \dots, 8$ and the N_b distribution for Gold $F(x)=x^3$ at $n=3,4,5,6$.

IMPORTANT LOGICAL POINT:

When $\max N_b \geq U_n$ (which happens for Gold x^3 at $n=4$ and $n=6$),

the oversized-fiber contrapositive does NOT apply. The script

prints an explicit warning in those cases, and confirms $N_1(F) > 0$

separately by direct pair count. Do not confuse:

- "all $N_b < U_n \Rightarrow N_1 > 0$ " (the useful contrapositive)
- "some $N_b \geq U_n \Rightarrow N_1 = 0$ " (FALSE; this direction does not hold)

NOTE: The "Extremal Σ_1 " column in the paper's threshold table contains a typographical error. The correct values (as computed by this script) are:

$n=3:$	704	$n=4:$	5888	$n=5:$	48128
$n=6:$	389120	$n=7:$	3129344	$n=8:$	25100288

The correct formula is $2^{\{2n\}} * (3 * 2^{\{n-1\}} - 1)$.

SECTION 6: WALSH TRACE-SLICE IDENTITY

Mathematical claim:

For APN F with $F(0)=0$, defining

$\Sigma_1(F) = \sum_{\{a \in GF(2^n), b \in GF(2^n)^*, \text{Tr}(b)=1\}} W_F(a,b)^3$

where $W_F(a,b) = \sum_x (-1)^{\{\text{Tr}(b \cdot F(x) + a \cdot x)\}}$ is the Walsh transform,

one has the exact identity:

$N_1(F) = 3 \cdot 2^n - 2 - 2^{\{1-2n\}} * \Sigma_1(F)$.

In particular, $N_1(F) = 0$ iff $\Sigma_1(F) = 2^{\{2n\}} * (3 \cdot 2^{\{n-1\}} - 1)$.

Method:

For $n = 3, 4, 5$ using Gold $F(x) = x^3$, the Walsh transform is computed by direct summation. Sigma_1 is accumulated, and the formula is compared to a direct count of N_1 .

Timing: $O(8^n)$ per dimension; only run for $n \leq 5$.

SECTION 7: FANO SUM IDENTITY

Mathematical claim:

For ANY F with $F(0)=0$ (not necessarily APN) and any three linearly independent elements u, v, w in $GF(2^n)$, the sum of the labels on all 7 lines of the Fano plane $PG(\langle u, v, w \rangle)$ equals the third-order derivative:

$$\sum_{l \in PG(\langle u, v, w \rangle)} \lambda_F(l) = D_u D_v D_w F(0)$$

where $D_u D_v D_w F(0) = F(u+v+w) + F(u+v) + F(u+w) + F(v+w) + F(u) + F(v) + F(w).$

Proof:

Each of the 7 nonzero points of the Fano plane lies on exactly 3

of the 7 lines. Summing over all lines:

$$\begin{aligned} \text{sum labels} &= \sum_p (\# \text{ lines through } p) * F(p) \\ &= 3 * \sum_p F(p) = \sum_p F(p) \quad (\text{in} \end{aligned}$$

characteristic 2)

and $\sum_p F(p)$ over the 7 Fano points is $D_u D_v D_w F(0)$ by definition.

Method: 200 random (F, u, v, w) trials per dimension for $n = 3, 4, 5, 6$.

SECTION 8: QUADRATIC FANO CONSTRAINT $B = C + D$

Mathematical claim:

For quadratic APN F (algebraic degree 2), every third-order derivative $D_u D_v D_w F(0)$ vanishes. Therefore the Fano sum identity gives sum of 7 labels = 0.

Writing the four independent Fano labels as A, B, C, D (as in the Fano-plane normal form of the paper), the seven labels are

$A, B, C, D, A+B+C, A+B+D, A+C+D.$

Their sum in characteristic 2 is $B + C + D$ (since A appears 4 times,

B, C, D each appear 3 times). Setting this to 0 gives $B = C + D.$

Illustrated with: Gold $F(x) = x^3$, which has algebraic degree 2

over $GF(2^n)$ for all $n \geq 2.$

Dimensions tested: $n = 3, 4, 5, 6.$

IMPLEMENTATION DETAILS

GF(2^n) arithmetic:

Elements are integers $0..2^n-1.$ Addition is XOR.

Multiplication

uses the shift-and-XOR algorithm with a fixed irreducible polynomial per $n.$

Projective line enumeration:

A line of $PG(n-1,2)$ is an unordered triple $\{a, b, c\}$ of nonzero

elements with $a \text{ XOR } b \text{ XOR } c = 0.$ The canonical form $a < b < c$

(equivalently: iterate over $a < b$, accept only when $c = a \text{ XOR } b > b$)

counts each line exactly once.

WARNING: iterating over pairs (a, b) with only $a < b$ and $c = a \text{ XOR } b$

counts each line THREE times (once per choice of which

two of the
 three points form the pair). The condition $c > b$ is
 essential.

APN testing: $O(4^n)$ per function.

$\emptyset$ -APN testing: $O(4^n)$ per function.

Walsh transform: $O(8^n)$ total for Sigma_1; only run for $n \leq 5$.

EXPORTED FUNCTIONS (for use in other scripts)

<code>gf_mul(a, b, poly, n)</code>	multiply in $GF(2^n)$
<code>gf_pow(a, e, poly, n)</code>	a^e in $GF(2^n)$
<code>is_apn(F, n)</code>	True iff F is APN
<code>is_0apn(F, n)</code>	True iff F is $\emptyset$ -APN
<code>line_label_set(F, n)</code>	(set_of_labels,
<code>count_of_zero_labels)</code>	
<code>line_label_count(F, n)</code>	dict {label:
<code>number_of_lines}</code>	
<code>Nb_direct(F, n, b)</code>	$N_b(F)$ by direct pair
<code>enumeration</code>	
<code>find_apn_monomials(n, poly)</code>	list of APN exponents d

CONTACT

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